1. Introduction to Control Systems

Control system = an interconnection of components forming a system configuration which gives a desired system response. In simple words: it **controls the output** of a process by manipulating the input so that output follows a desired value (reference).

Basic terms (write these in every answer)

- **System** – a combination of components acting together to perform a function.
- **Plant / Process** – the physical object to be controlled (motor, furnace, aircraft).
- **Input (reference) $r(t)$** – the desired value; **Output $c(t)$** – the actual controlled variable.
- **Controller** – generates the control signal from the error. **Actuator** – drives the plant.
- **Sensor / Feedback element** – measures output and sends it back. **Error $e(t) = r(t) - b(t)$.**
- **Disturbance** – unwanted signal that affects output (load change, wind, noise).

Examples

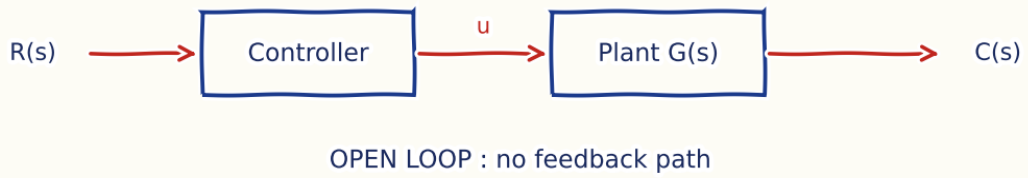
Room air-conditioner (thermostat), water-level control in a tank, speed control of a DC motor, automatic washing machine (open loop), traffic light (open loop), missile guidance, robot arm, cruise control in a car.

> ★ **Aim of a control system: (i) accuracy (ii) stability (iii) fast response (iv) rejection of disturbances (v) low sensitivity to parameter variations.**

2. Open Loop and Closed Loop Systems

Open loop system

*The control action is **independent of the output**. There is no feedback; output is neither measured nor compared with the input.*

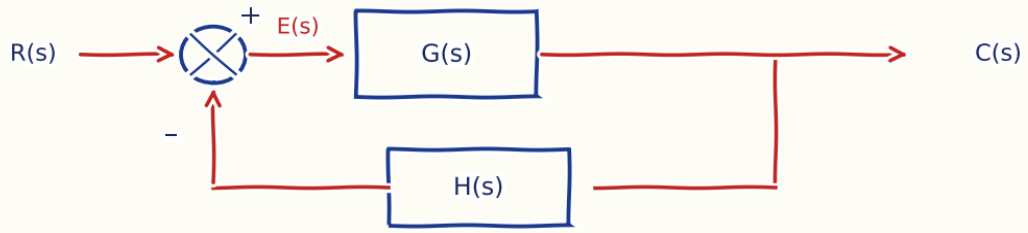


OPEN LOOP : no feedback path

***Example:** traffic signal (timer based), automatic washing machine, electric toaster, immersion rod with timer.*

Closed loop (feedback) system

*The control action depends on the output. Output is measured, fed back and compared with the reference. The difference (error) actuates the controller to reduce the error. Also called **feedback control system**.*



CLOSED LOOP : feedback H(s)

***Example:** room temperature control by thermostat, human being (eye–hand), speed governor, servo-mechanism, autopilot.*

Closed loop transfer function (negative feedback): $C(s)/R(s) = G(s) / [1 + G(s)H(s)]$

Positive feedback: $C(s)/R(s) = G(s) / [1 - G(s)H(s)]$

Differences — Open loop vs Closed loop (most asked 5M/10M)

S.No	Open Loop	Closed Loop
1	No feedback; output has no effect on control action	Feedback present; output affects control action
2	Simple, cheap, easy to design	Complex, costly, more components
3	Less accurate, accuracy depends on calibration	Highly accurate (error is corrected)
4	Stable (no stability problem)	May become unstable if gain is high
5	Sensitive to parameter variation & disturbance	Less sensitive to disturbance and parameter changes
6	Bandwidth is small	Bandwidth is large
7	Reliable for known, constant conditions	Used when conditions vary
8	Eg: traffic signal, toaster	Eg: AC thermostat, servo motor

3. Classification of Control Systems

Basis	Types	Remarks
Feedback	Open loop / Closed loop	With or without feedback path
Linearity	Linear / Non-linear	Linear obeys superposition: additivity + homogeneity
Time variation	LTI / LTV	LTI: parameters constant with time; LTV: vary with time (rocket mass)
Signal	Continuous-time / Discrete-time	Discrete uses sampled data, z-transform
Inputs–outputs	SISO / MIMO	Single / multiple input–output
Parameters	Lumped / Distributed	ODE vs PDE (transmission line)
Nature	Deterministic / Stochastic	Predictable vs random response
Purpose	Regulator / Servo / Process	Regulator keeps output constant; servo follows a varying reference

Superposition principle: *if* $r_1 \rightarrow c_1$ *and* $r_2 \rightarrow c_2$ *then* $(a\, r_1 + b\, r_2) \rightarrow (a\, c_1 + b\, c_2)$. *Systems with saturation, dead-zone, backlash, relay, friction are* ***non-linear***.

4. Feedback Characteristics and Effects of Feedback

Consider a negative feedback system with forward gain G and feedback gain H. Overall gain:

$$M = C/R = G / (1 + GH)$$

(a) Effect on overall gain

$|1 + GH| > 1$ *for negative feedback, hence overall gain is* ***reduced*** *by the factor* $(1 + GH)$. *For positive feedback gain increases but stability is lost.*

(b) Effect on sensitivity (parameter variation)

Sensitivity of M with respect to G = fractional change in M per fractional change in G:

$$S^{M}_{G} = (\partial M/M)/(\partial G/G) = 1 / (1 + GH)$$
$$S^{M}_{H} = - GH / (1 + GH)$$

*If $GH >> 1 : S^{M}_{G} \approx 0 \rightarrow$ the system is almost **insensitive** to variations in G (plant) but sensitive to H, so use a precise feedback element (sensor).*

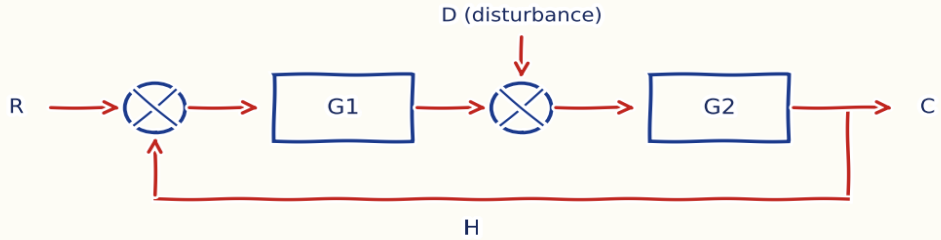
(c) Effect on stability

Feedback can make a stable open loop unstable (gain too high, phase lag at high frequency). The characteristic equation 1 + G(s)H(s) = 0 decides stability. Right choice of gain keeps system stable, and feedback can also stabilise an unstable plant.

(d) Effect on bandwidth

*Feedback **increases bandwidth** (gain × bandwidth ≈ constant), giving faster response (smaller rise time).*

(e) Effect on disturbance and noise



$$C = [G1G2 /(1+G1G2H)]\ R + [G2 /(1+G1G2H)]\ D$$

*With large G1H the effect of disturbance D on the output is reduced by (1 + G1G2H), whereas in open loop the disturbance appears fully at the output. Hence feedback gives **disturbance rejection** and reduces noise in the loop.*

(f) Other effects

- Reduces **steady-state error** (improves accuracy).
- Reduces non-linear distortion and effects of parameter drift.
- Improves transient response (can be tuned using gain), but may introduce oscillations.
- Increases complexity and cost; needs a sensor.

> ★ **Short answer: Feedback reduces gain, sensitivity, noise and error; it increases bandwidth, accuracy and disturbance rejection; but may reduce stability.**

5. Mathematical Models of Physical Systems

To analyse a system we first obtain its **mathematical model**, a set of equations that represents its dynamics. Methods: (i) differential equations (time domain) (ii) transfer function (s-domain) (iii) state-space model (Unit-V).

Differential equation model

Use physical laws: Newton's law (mechanical), Kirchhoff's laws (electrical), thermal and fluid laws. A general n^{th} order LTI system:

$$a_n \, d^n c/dt^n + \ldots + a_1 \, dc/dt + a_0 \, c = b_m \, d^m r/dt^m + \ldots + b_0 \, r$$

Transfer function (T.F.)

Definition: the ratio of the Laplace transform of output to the Laplace transform of input, with **all initial conditions zero**, for an LTI system.

$$G(s) = C(s)/R(s) = (b_m s^m + \ldots + b_0) \, / \, (a_n s^n + \ldots + a_0)$$

- Roots of numerator = **zeros**, roots of denominator = **poles**; denominator = characteristic polynomial; order n = highest power of s in denominator.
- T.F. is independent of input; applicable only to LTI systems; gives no information about the physical structure.

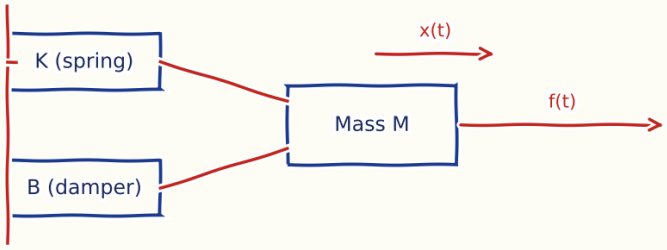
- T.F. = Laplace transform of the **impulse response**.
- Advantages: algebraic manipulation, easy stability and response analysis. Limitation: zero initial conditions, not for non-linear / time-varying / MIMO (conveniently).

Laplace pair	f(t)	F(s)
Unit step	$u(t)$	$1/s$
Unit ramp	t	$1/s^2$
Impulse	$\delta(t)$	1
Exponential	e^{-at}	$1/(s+a)$
Sine / Cosine	$\sin \omega t$ / $\cos \omega t$	$\omega/(s^2+\omega^2)$, $s/(s^2+\omega^2)$
Derivative	df/dt	$sF(s) - f(0)$
Integral	$\int f\, dt$	$F(s)/s$

6. Translational Mechanical Systems

Three basic elements (Newton's 2nd law: $\Sigma F = M\, a$):

Element	Force relation	Laplace form
Mass M	$f = M\, d^2x/dt^2$	$F(s) = M\, s^2\, X(s)$
Viscous friction (damper) B	$f = B\, dx/dt$	$F(s) = B\, s\, X(s)$
Spring K	$f = K\, x$	$F(s) = K\, X(s)$



Example 1: Mass–spring–damper

Applied force $f(t) = M\,x'' + B\,x' + K\,x$. Taking Laplace transform with zero initial conditions:

$$X(s)/F(s) = 1 / (M s^2 + B s + K)$$

Writing in standard form: $\omega_n = \sqrt{(K/M)}$, $\zeta = B / (2\sqrt{(KM)})$.

Example 2: Two masses (M1, M2) with springs K1, K2 and dampers B1, B2

Write one equation per mass (node), the relative motion terms use $(x1 - x2)$:

$$M1\,x1'' + B1\,x1' + K1\,x1 + B2\,(x1' - x2') + K2\,(x1 - x2) = f(t)$$

$$M2\,x2'' + B2\,(x2' - x1') + K2\,(x2 - x1) + B3\,x2' = 0$$

★ Procedure: (1) assume direction of x (2) draw free-body diagram for each mass (3) write $\Sigma F = Ma$ (4) Laplace transform (5) solve for X(s)/F(s) using Cramer's rule.

7. Rotational Mechanical Systems

Element	Torque relation	Laplace form
Moment of inertia J	$T = J\,d^2\theta/dt^2$	$T(s) = J\,s^2\,\theta(s)$
Rotational friction B	$T = B\,d\theta/dt$	$T(s) = B\,s\,\theta(s)$
Torsional spring K	$T = K\,\theta$	$T(s) = K\,\theta(s)$

Newton's law for rotation: $\Sigma\tau = J\,\alpha$. Example: a disc with inertia J, shaft stiffness K, bearing friction B and applied torque T(t):

$$T(t) = J\,\theta'' + B\,\theta' + K\,\theta \Rightarrow \theta(s)/T(s) = 1 / (J s^2 + B s + K)$$

For gear trains, reflect parameters to one shaft: $J_{eq} = J_1 + (N_1/N_2)^2 J_2$, $B_{eq} = B_1 + (N_1/N_2)^2 B_2$, where N is number of teeth $(\theta_2/\theta_1 = N_1/N_2)$.

8. Electrical–Mechanical Analogies

Mechanical (translational)	Force–Voltage analogy (series RLC)	Force–Current analogy (parallel RLC)
Force f	Voltage e	Current i
Mass M	Inductance L	Capacitance C
Friction B	Resistance R	Conductance $1/R$
Spring K	Elastance $1/C$	$1/L$
Displacement x	Charge q	Flux ψ
Velocity v	Current i	Voltage e

Rotational: torque T ↔ voltage (or current), J ↔ L (or C), B ↔ R (or $1/R$), K ↔ $1/C$ (or $1/L$), θ ↔ q. Analogy lets us study a mechanical system on an electrical circuit (and vice-versa). Series mechanical connection ↔ parallel electrical in F–I analogy.

9. Transfer Function of DC Servo Motor

A DC servo motor is a DC motor used in a feedback position/speed control system; it has low inertia, high torque/inertia ratio and linear torque–speed behaviour. Two types: ***armature controlled*** *and* ***field controlled****.*

(A) Armature controlled DC motor (field current constant)

Symbols: R_a, L_a armature resistance & inductance; e_a applied voltage; e_b back emf; K_t torque constant; K_b back-emf constant; J, B motor inertia, friction; θ shaft angle.

$$\textbf{Armature circuit: } e_a = i_a R_a + L_a \frac{di_a}{dt} + e_b$$

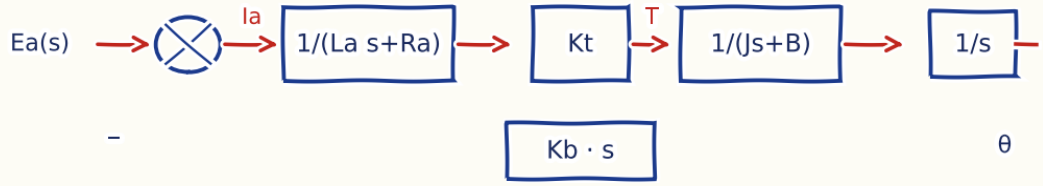
$$\textbf{Back emf: } e_b = K_b \frac{d\theta}{dt}$$

$$\textbf{Torque: } T = K_t i_a$$

$$\textbf{Mechanical: } T = J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt}$$

Laplace transform (zero initial conditions):

$$E_a - K_b s\theta = (R_a + L_a s) I_a \; ; \; K_t I_a = (Js^2 + Bs)\theta$$



$$\theta(s)/E_a(s) = K_t / \{ s [(R_a + L_a s)(Js + B) + K_t K_b] \}$$

$$\text{If } L_a \approx 0 : \theta(s)/E_a(s) = K_m / [s (\tau_m s + 1)]$$

$$K_m = K_t / (R_a B + K_t K_b) , \tau_m = R_a J / (R_a B + K_t K_b)$$

*K_m = motor gain constant, τ_m = motor time constant. Note: back emf K_b acts like **internal feedback** that adds to damping (effective friction $B + K_t K_b / R_a$).*

(B) Field controlled DC motor (armature current constant)

Field circuit: $e_f = R_f i_f + L_f di_f/dt$. Torque $T = K_f i_f$ (flux $\propto i_f$, I_a constant), $T = J\theta'' + B\theta'$.

$$\theta(s)/E_f(s) = K_f / \{ s (L_f s + R_f)(Js + B) \} \text{ (third order)}$$

$$= K_m / \{ s (\tau_f s + 1)(\tau_m s + 1) \} , \tau_f = L_f/R_f , \tau_m = J/B$$

Point	Armature controlled	Field controlled
Order of T.F.	2nd (La neglected)	3rd
Constant quantity	Field current	Armature current
Efficiency / use	Better, widely used	Poor, small motors only
Inherent damping	Yes (back emf)	No

10. Transfer Function of AC Servo Motor

A two-phase induction motor with ***two stator windings*** *90° apart: a* ***reference winding*** *(fixed ac voltage) and a* ***control winding*** *(variable ac voltage* e_c*, 90° out of phase). Rotor has high resistance so the torque–speed curve is linear with negative slope (good damping, no run-away).*

Linearised torque equation around operating point:

$$T_m = K_1 e_c - K_2 \, d\theta/dt \; (K_1 = \text{control phase constant}, K_2 = \text{slope of } T\text{–}\omega \text{ curve})$$

$$T_m = J \, d^2\theta/dt^2 + B \, d\theta/dt$$

$$\square(s)/E_c(s) = K_1 / \{ s [Js + B + K_2] \} = K_m / [s (\tau_m s + 1)]$$

$$K_m = K_1 / (B + K_2) \, , \; \tau_m = J / (B + K_2)$$

- *Used where low power (< 1 hp) is enough: instruments, radar, small robots.*
- *Advantages: no commutator/brush, rugged, low maintenance.*
  Disadvantage: lower efficiency, non-linear characteristics, low torque.

> ★ ***AC servo vs DC servo: AC – brushless, low power, 50/400 Hz, lower efficiency; DC – brushed, high power, more efficient, commutator maintenance required.***

11. Synchro Transmitter and Receiver

A ***synchro*** *is an electromagnetic transducer (rotary transformer) that converts angular position of a shaft into an electrical signal; used as an* ***error detector*** *in servo systems. A synchro pair (transmitter + control transformer/receiver) measures angular difference between two shafts.*

Synchro transmitter (Tx)

Construction: a **rotor** (dumb-bell, single winding on slip rings, excited by a.c. voltage $V_r \sin \omega t$) and a **stator** with three identical star-connected windings S1, S2, S3 spaced 120° apart. The rotor flux induces emfs in the stator windings, magnitude depending on rotor angle θ.

$$e_{S1n} = K V_r \sin \omega t \cos \theta$$
$$e_{S2n} = K V_r \sin \omega t \cos(\theta - 120°)$$
$$e_{S3n} = K V_r \sin \omega t \cos(\theta + 120°)$$

Line voltages: $e_{S1S2} = \sqrt{3}\, K V_r \sin \omega t \sin(\theta + 240°)$

$e_{S2S3} = \sqrt{3}\, K V_r \sin \omega t \sin \theta$, $e_{S3S1} = \sqrt{3}\, K V_r \sin \omega t \sin(\theta + 120°)$

Maximum line voltage ($\sqrt{3}\, K V_r$) is obtained when rotor is at 90° with respect to the stator coil axis; at θ = 0°, 120°, 240° one of the line voltages is zero. These 3 line voltages give a **unique** set for every angle → position information.

Synchro receiver / control transformer (CT) as error detector

The stator of the control transformer is connected to the Tx stator. Rotor of CT is not excited; the voltage across rotor terminals is the output:

$$e_0(t) = K' \sin \omega t \cdot \sin(\theta_t - \theta_r) \approx K' (\theta_t - \theta_r) \sin \omega t \text{ for small error}$$

So output is a suppressed-carrier amplitude-modulated signal proportional to the **error** $\theta_e = \theta_t - \theta_r$. It is demodulated and amplified to drive a servo motor. Transfer function of the error detector:

$$E_0(s)/\square_e(s) = K_s \text{ (sensitivity of synchro, volt/rad} \approx \text{volts per degree)}$$

Torque synchro (receiver) type: the rotor of receiver is also excited; Tx and Rx stators joined; if rotor positions differ a torque acts to align them. Equilibrium (zero torque) when both rotor angles are equal. Receiver rotor

has a damper to avoid oscillations.

- *Advantages: robust, no contact (except slip rings), accurate (± 0.1°), operates over long distances.*
- *Applications: remote position indication, radar antenna, gun-turret, robot arms, machine tools.*

12. Quick Revision / Important Questions

- *Define control system; differentiate open and closed loop with examples. (5M)*
- *Explain the effects of feedback on gain, sensitivity, stability, bandwidth, noise. (10M)*
- *Derive T.F. of armature controlled DC servo motor with block diagram. (10M)*
- *Derive the T.F. of field controlled DC motor. Compare with armature control. (10M)*
- *Write the differential equations for the given mechanical system and draw force–voltage and force–current analogous circuits. (10M)*
- *Explain the construction and working of synchro transmitter and receiver; derive the line voltages. (10M)*
- *Derive the T.F. of AC servo motor from its torque–speed characteristic. (5M)*
- *Define transfer function; list its properties/limitations. (2M/5M)*

$$C/R = G/(1+GH) \mid S^{M}_{G} = 1/(1+GH) \mid X/F = 1/(Ms^2+Bs+K)$$

$$\text{Armature motor: } K_t/\{s[(R_a+L_a s)(Js+B)+K_t K_b]\} \mid \text{AC servo: } K_m/[s(\tau_m s+1)]$$

★ ***Exam tip: always draw the block diagram / free-body diagram first, state assumptions (zero initial conditions, L_a neglected) and box the final transfer function.***